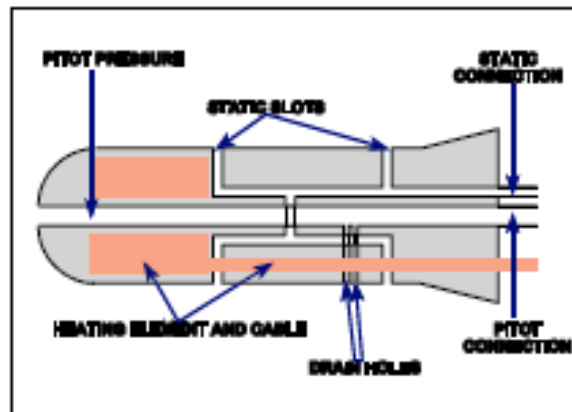


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Pressure Heads

- An aircraft at all times and on all parts is subjected to normal atmospheric pressure which bears equally on all parts of the aircraft. This ambient pressure is known as **Static pressure**.
- An aircraft in flight, while still subject to the static pressure at its height, experiences an additional pressure on the leading edges due to the resistance of the air to the aircraft's movement. This additional pressure is **Dynamic pressure**.
- The leading edges, therefore, encounter a total pressure consisting of static pressure plus dynamic pressure. This total pressure is known as **Pitot pressure**.
- The altimeter and vertical speed indicator, operate solely on static pressure.
- The airspeed indicator and machmeter utilise both static and pitot pressures.



Combined Pitot/Static Pressure Head

- **Position error** is due to turbulent airflow in the region of the Pitot/Static Head.
- Approximately 95% of the position error associated with a combined pressure head is produced by turbulence about the static head. Because of this, the Static Vent, was introduced and about 90% of the combined pressure head position error is eliminated by use of a separate pitot head and static vent.
- **Manoeuvre-induced errors** are caused by short-term fluctuations of pressure at the static vents and delays in the associated pipelines transmitting pressure changes to the instruments.
- An emergency static source is normally provided in the event of the static head/vents becoming blocked. The emergency static source may be to the outside of the aircraft or from inside the cabin (in unpressurised aircraft only).

QUESTIONS

1. **A pitot head is used to measure:**
 - a. dynamic minus static pressure.
 - b. static plus dynamic pressure.
 - c. static pressure.
 - d. dynamic pressure.
2. **A static vent is used to measure:**
 - a. dynamic pressure minus pitot excess.
 - b. dynamic pressure plus pitot excess.
 - c. atmospheric pressure.
 - d. pitot excess pressure.
3. **A pressure head is subject to the following errors:**
 - a. position, manoeuvre induced, temperature.
 - b. position, manoeuvre induced.
 - c. position, manoeuvre induced, density.
 - d. position, manoeuvre induced, instrument.
4. **Turbulent flow around a pressure head will cause:**
 - a. density error.
 - b. 95% increase in manoeuvre induced error.
 - c. an increase in the dynamic pressure.
 - d. 95% of pressure error.
5. **Manoeuvre induced error:**
 - a. is caused by transient pressure changes at static vents.
 - b. is likely to be greatest when yawing after engine failure.
 - c. is combined with instrument and position error on a correction card.
 - d. lasts for only a short time at high altitude.
6. **Position error:**
 - a. may be reduced by the fitting of static vents.
 - b. will usually decrease with an increase in altitude.
 - c. will depend solely on the attitude of the aircraft.
 - d. will usually decrease as the aircraft approaches the speed of sound.
7. **Pressure heads supply data to the following instruments:**
 - a. air data computers, compasses, altimeters, and ASI's.
 - b. standby instruments only, when air data computers fitted.
 - c. altimeters, ASI's, VSI's, machmeters, air data computers.
 - d. all the above plus air driven gyros.

8. **Static vents are usually fitted to both sides of the aircraft fuselage. This will:**
- a. reduce the position error.
 - b. balance out errors caused by side slipping or yawing.
 - c. require a calibration card for each static vent.
 - d. enable a greater number of instruments to be fitted.
9. **Which of the following instruments require inputs of both pitot and static pressure:**
- a. airspeed indicator, machmeter and vertical speed indicator.
 - b. airspeed indicator, vertical speed indicator, altimeter.
 - c. airspeed indicator only.
 - d. airspeed indicator and machmeter.
10. **Where an alternate static source is fitted, use of this source usually leads to:**
- a. a temporary increase in lag error.
 - b. a lower pressure error than with normal sources.
 - c. an increase in position error.
 - d. no change in position error.

ANSWERS

1	2	3	4	5	6	7	8	9	10
B	C	D	D	A	A	C	B	D	C

Air Temperature Measurement

The measurement of air temperature is important to a pilot, not just to establish the likelihood of engine or airframe icing, but also in connection with many other aspects of aircraft performance, and flight planning.

The Effect of Compressibility-

As aircraft speed increases, the air close up to the aircraft becomes compressed. Due to this compression the air becomes heated. This means that the temperature sensed will be too warm and not representative of the actual air temperature outside of the aircraft.

- **Static Air Temperature (SAT)** is the temperature of the undisturbed air through which the aircraft is about to fly.
- **Total Air Temperature (TAT)** is the maximum temperature attainable by the air when brought to rest, adiabatically.
- The increase of air temperature at higher speeds as a result of the adiabatic compression of the air is known as the **Ram Rise**.
- The percentage of the “Ram Rise” sensed, and recovered, by a TAT probe is termed the Recovery Factor (Kr).

A Total Air Temperature Probe-

- The probe is in the form of a small strut and air intake made of nickel-plated beryllium copper which provides good thermal conductivity and strength.
- It is secured at a pre-determined location in the front fuselage section of an aircraft outside of any boundary layer.
- A pure platinum wire resistance-type sensing element is used. The probe has an almost negligible time-lag, and a high recovery factor of approximately 1.00.



The relationship between heating error, SAT and TAT is shown:-

$$\text{SAT} = \text{TAT} - \left(\frac{V}{100}\right)^2$$

Where:

V is the true airspeed in knots

SAT is the Static Air Temperature

TAT is the Total Air Temperature (TAT = SAT + Ram Rise)

If Mach no. is given -

$$\text{SAT} = \frac{\text{TAT}}{1 + 0.2 \text{ Kr } M^2}$$

where:

SAT = Correct outside air temperature

TAT = Indicated outside air temperature

M = Mach number

Kr = Recovery factor of the temperature bulb

Questions

1. **Flying at high speed at high altitude, the difference between ram air temperature and static air temperature is:**
 - a. likely to be less than when flying low and slow.
 - b. due to adiabatic cooling.
 - c. due to adiabatic warming.
 - d. proportional to the square of the absolute temperature.
2. **At a true airspeed of 500 knots, a ram rise of air temperature can be expected of:**
 - a. 50°C
 - b. 25°C
 - c. 5°C
 - d. 16°C
3. **Total Air Temperature is:**
 - a. the maximum temperature attainable by the air when brought to rest, adiabatically.
 - b. the temperature indicated on the air temperature thermometer plus the ram rise.
 - c. the static air temperature minus the recovery factor.
 - d. the recovery factor plus the ram rise.
4. **The difference between static air temperature and total air temperature is known as:**
 - a. corrected outside air temperature.
 - b. the ram rise.
 - c. the recovery factor.
 - d. hot ramp radiation.

Answers

1	2	3	4
C	B	A	B

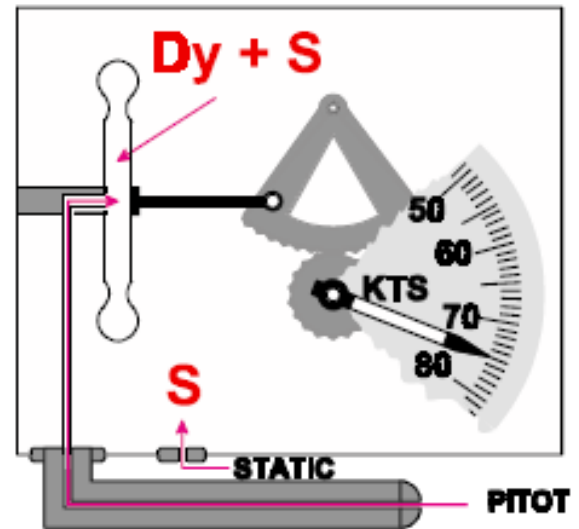
Air Speed Indicator

- The ASI measures airspeed by measuring dynamic pressure, displaying the result (usually in knots) on a suitably calibrated scale.
- The pitot head senses pitot pressure and the static/vent senses static pressure.
- These two pressures are fed to the airspeed indicator, a differential pressure gauge, which measures their difference (the dynamic pressure).
- Now dynamic pressure is a measure of airspeed, because:-

$$\text{Dynamic Pressure} = \frac{1}{2} \rho V^2$$

Construction -

- The static pressure is fed into the hermetically-sealed instrument case, the pitot pressure being piped to a thin metal capsule capable of expansion and contraction.
- Note that the pressure differential between the inside and outside of the capsule is $(Dy + S) - S$ which is Dy .
- Expansion or contraction on the capsule will therefore be proportional to the changes in dynamic pressure produced by changes of airspeed.
- The capsule movements are transmitted by a temperature-compensated magnifying linkage to the pointer indicating airspeed on the face of the ASI.



Calibration -

The ASI is calibrated to read true airspeed for the air density of 1.225 kg/m^3 which would be produced by the ISA MSL pressure of 1013.25 hPa and temperature $+15^\circ\text{C}$ (dry air conditions). No allowance is made in the calibration for the change in density which occurs with change of altitude.

ASI Errors -

1. **Instrument Error** - Manufacturing imperfections.
2. **Position/Pressure Error** – Arises due to incorrect pressure entering the vent. It is maximum at high AOA and low airspeeds. Maximum Position error means more static is entering.

Manoeuvre-Induced Errors - These are associated chiefly with manoeuvres involving change in angle of attack, giving transient errors and a lag in the indication of changes in airspeed.

4. Compressibility Error –

- It occurs when TAS is more than 300kts.
- At high altitudes, less dense air is easily compressed than denser air at sea level resulting in greater dynamic pressure
- This causes **ASI to Over-Read**.
- This error increases directly with both Airspeed and Altitude.

5. Density Error –

- It is the biggest error of ASI.
- If the air round the aircraft is not calibrated at ISA density of 1.225kg/m^3 , the ASI cannot correctly indicate TAS.
- The ASI under-reads the true speed at altitude.
- If below MSL, the ASI will over read the true speed.

Summary of Errors –

- IAS (indicated air speed) $\pm$ P and I correction = CAS
- CAS + correction for density error (nearly always +) = TAS

Air Speed Definitions-

IAS (Indicated Air Speed)	Indicated reading on instrument
CAS(Calibrated Air Speed)	IAS $\pm$ correction for instrument & position error
EAS(Equivalent Air Speed)	CAS $\pm$ compressibility corrections
TAS(True Air Speed)	CAS $\pm$ density error $\pm$ compressibility error or EAS $\pm$ density error

More ASI Definitions-

V_{SO} = The stall speed or the minimum steady flight speed in the landing configuration.

V_{S1} = The stall speed or the minimum steady flight speed in a specified configuration.

V_{FE} = The maximum Flap Extension speed

V_{NO} = The maximum normal operating limit speed.

V_{NE} = The Never Exceed speed

V_{LO} = The maximum Landing Gear Operation speed (up or down).

V_{MO} = The maximum operating speed.

V_{LE} = The maximum speed Landing Gear Extended speed

V_{YSE} = Best rate of climb when Single Engine (2 eng a/c)

ASI Color Coding –

- **The White Arc** - V_{SO} to V_{FE} . (The Flap Operating Range)
- **The Green Arc**- V_{SI} to V_{NO} (The Normal Operating Speed Range)
- **The Yellow Arc**- V_{NO} to V_{NE} (The Caution Range)
- **Red Radial Line** - V_{NE}
- **Blue Radial Line** – V_{YSE}
- **A Lower Red Radial Line**- V_{MCA}



Note-

$$TAS = CAS + (1.75\% \text{ of CAS per } 1,000\text{ft of Altitude})$$

Eg - For a CAS of 100 kt at 10,000 ft: $TAS = CAS + (1.75 \times 100 \times 10) = 117.5 \text{ kt}$

Blockages-

- Pitot Head** - If the pitot head becomes blocked, the ASI reading will remain unchanged. However, the capsule will behave as a barometer (Altimeter) producing an indication of increase of speed if the a/c climbs & decrease of speed if the a/c descends.
- Static** – If the static is blocked, ASI will over-read if A/c descends at lower altitudes (This could be dangerous in that the aircraft is nearer the stall than the ASI is indicating). Similarly, it will under read when climbing at high altitudes.

Leaks –

- Pitot Head** – A leak in pitot causes ASI to under read.
- Static** – In unpressurised a/c – ASI over reads.
In pressurised a/c – ASI under reads.
(Outside air pressure is higher than static)

Static Balancing –

- In bigger a/c static ports are placed on both sides of the fuselage. This is called Static Balancing.
- It helps in reducing position error when a/c is side slipping in a turn.
- Whenever A/c sideslips the static port on the other side helps balancing the extra pressure sensed on one side.
- If a/c sideslips towards the blocked static port, ASI over reads. If a/c sideslips towards open port assuming other side is blocked, ASI will under read.

QUESTIONS

1. **If the pitot line to an ASI becomes totally blocked during a climb, the ASI reading will:**
 - a. decrease, no matter what the actual airspeed is.
 - b. increase, no matter what the actual airspeed is.
 - c. progressively under indicate the value of the airspeed.
 - d. stick at the airspeed showing at the moment of blockage.
2. **Compressibility error:**
 - a. causes overreading of the ASI at all levels whenever TAS exceeds 300 kt.
 - b. causes overreading of the ASI whenever CAS exceeds 300 kt.
 - c. is calibrated out of the ASI at MSL ISA conditions and only causes over reading when air density is decreased and CAS exceeds 300 kt.
 - d. is calibrated out of the ASI at MSL ISA conditions and only causes over reading when air density is decreased and TAS exceeds 300 kt.
3. **If the static line to the ASI becomes blocked during a long descent, a dangerous situation could arise due to the ASI:**
 - a. overreading, this indicated speed falsely showing the aircraft to be further from the stalling speed than it actually is.
 - b. underreading, this indicated speed falsely showing the aircraft to be closer to the stalling speed than it actually is.
 - c. underreading, this indicated speed possibly leading to the operation of flaps and/or landing gear at speeds in excess of safety speeds.
 - d. overreading, this indicated speed possibly leading to the operation of flaps and/or landing gear at speeds in excess of safety speeds.
4. **An aircraft maintaining a constant CAS and altitude is flying from a cold airmass into warmer air. The effect of the change of temperature on the speed will be:**
 - a. CAS will decrease.
 - b. EAS will increase.
 - c. TAS will increase.
 - d. TAS will decrease.
5. **The airspeed indicator is calibrated to:**
 - a. conditions of the International Standard Atmosphere.
 - b. conditions of the International Standard Atmosphere at MSL.
 - c. an air density of 1013.25 gm/m³.
 - d. indicate correctly in any atmosphere.
6. **Dynamic pressure is equal to:**

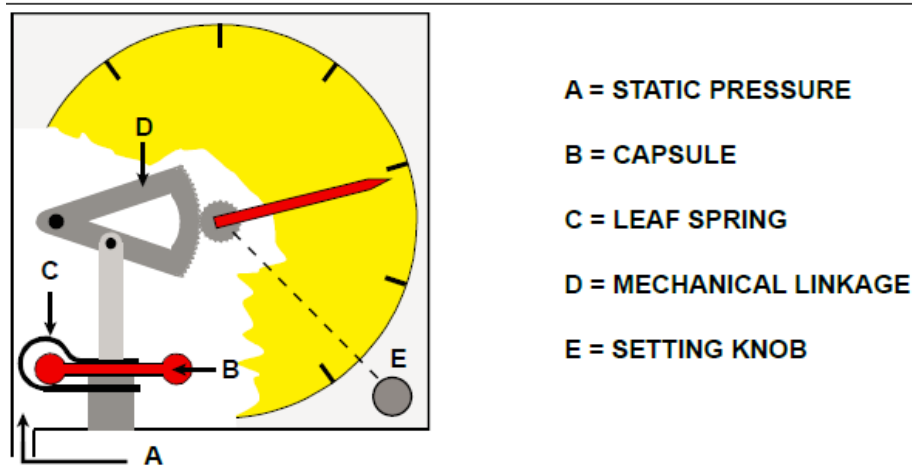
- a. $\frac{1}{2} vQ^2$
 b. $\frac{1}{2} Qv^2$
 c. $(\frac{1}{2} Qv)^2$
 d. $\frac{1}{2} (Qv)^2$
7. **Excluding blockages, the full list of errors of the ASI is:**
 a. instrument error, position error, density error, manoeuvre induced error.
 b. instrument error, position error, temperature error, compressibility error, manoeuvre induced error.
 c. instrument error, position error, barometric error, temperature error, lag, manoeuvre induced error.
 d. instrument error, position error, density error, compressibility error, manoeuvre induced error.
8. **Some ASIs have coloured arcs and lines marked on their dials. A yellow arc and a white arc indicate:**
 a. cautionary range and normal operating range.
 b. flap operating speed range and normal operating range.
 c. cautionary range and flap operating speed range.
 d. flap operating speed range and cautionary range.
9. **What will be the TAS if cruising altitude is 39 000 ft, temperature is ISA +5 and CAS 200 kt:**
 a. 388 kt
 b. 383 kt
 c. 364 kt
 d. 370 kt
10. **If the static line to the ASI becomes blocked during a climb, the ASI reading will:**
 a. increase, no matter what the actual airspeed is.
 b. progressively under indicate the value of airspeed.
 c. progressively over indicate the value of airspeed.
 d. stick at the airspeed showing at the moment of blockage.

Answers

1	2	3	4	5	6	7	8	9	10
B	D	A	C	B	B	D	C	B	B

Pressure Altimeter

The pressure altimeter is a sensitive pressure gauge calibrated to indicate height above selected datum level.



1. The simple altimeter – Construction –

- Pressure altimeter consist of thin metal capsule which is evacuated and sealed.
- Case is fed with static pressure from a/c static tube or vent.
- As height increases, static pressure decreases and the capsule expands under the control of a leaf spring.
- Conversely, decrease in the height compresses the capsule.
- A mechanical linkage magnifies the capsule expansion and converts it to a rotational movement of a single pointer over the height scale.

2. Sensitive Altimeter –

- The sensitive altimeter employs a minimum of two aneroid capsules. This provides for a more accurate measurement of pressure and provides more power to drive the mechanical linkage.
- Jewelled bearings are fitted, reducing friction and the associated lag in indications.
- Within the mechanical linkage, a bi-metallic insert is fitted to compensate for temperature changes that could affect the movement.

3. Servo Assisted Altimeter –

- In this, instead of mechanical amplifications, the capsule assembly is made part of the Servo System.
- This servo-assistance not only gives the altimeter an increased operating range but also improves the instrument accuracy (Accuracy $\pm 60\text{ft}$).

- Electro-Magnetic Induction Pick-off device is used to virtually eliminate Lag Error.
- Function of vibrator is to reduce friction within the linkages.

Accuracy –

i. Simple Altimeter

Height (feet)	MSL	35,000
Accuracy (feet)	±100	± 1,000

ii. Sensitive Altimeter

Height (feet)	MSL	40,000
Accuracy (feet)	±70	±600

iii. Servo Altimeter

Height (feet)	MSL	40,000
Accuracy (feet)	±30	±100

Errors in Altimeter –

- 1. Instrument Error** – Due to manufacturing imperfections.
- 2. Lag Error** - The response of capsule or linkages to change of height is not instantaneous. This causes the altimeter to under-read in a climb and over-read in a descent. Lag error is virtually eliminated in Servo-Assisted Altimeters.
- 3. Position Error** - This is largely due to the inability to sense the true static pressure outside the aircraft.
- 4. Barometric (or Pressure) Error** – Occurs when actual datum pressure differs from ISA. Barometric error of roughly 30 feet per hPa will result. If pressure has fallen the altimeter over-reads and vice-versa.
- 5. Temperature Error** –
 - Occurs when actual temperature differs from ISA.
 - In a difference of 10°C from ISA an error of approximately 4% of indicated altitude will occur.
 - Altimeter over reads when temperature decreases and vice-versa.
 - In flight from HIGH TO LOW TEMPERATURE the altimeter would read HIGH.
(Dangerous)
- 6. Hysteresis Error** – The capsule suffers from hysteresis error which causes a lag in the instrument reading during climb or descent due to time passed over a given altitude.
- 7. Blockage** - If the static source becomes blocked, the pressure within the casing will remain constant. Altimeter will not register any change in height - the height at which the blockage occurred will still be indicated regardless of any climb or descent.

8. Leaks –

- Static line leak in a pressurised aircraft, the altimeter will show the (lower) cabin altitude rather than aircraft altitude.
- A leak in the static line within an unpressurised aircraft will normally result in the altimeter over reading, as static pressure inside the a/c is slightly less than local atmospheric static pressure. So a leak in static line will allow less pressure to enter static lines causing altimeter to over read.

Altimetry Definitions –

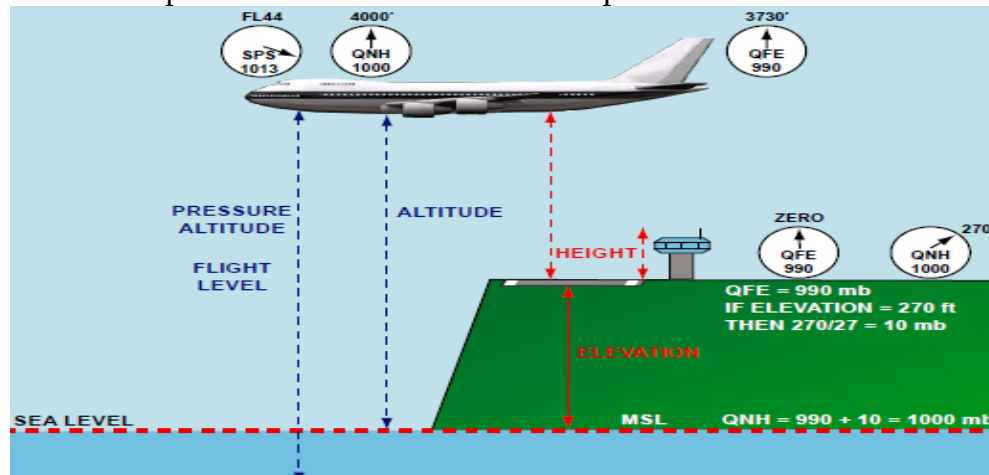
Height - The vertical distance of a level, point or object considered as a point, measured from a specified datum.

Altitude - The vertical distance of a level, point or object considered as a point, measured from MSL.

Elevation – The vertical distance of a level or a point on or affixed to the surface of the earth, measured from MSL.

Cruising Level - This is a generic term describing vertical position for a significant portion of the flight and can be a height, altitude, or flight level depending on the altimeter setting procedure in force.

Pressure Altitude - The atmospheric pressure expressed as in terms of altitude which corresponds to that pressure in the standard atmosphere.



Pressure Definitions-

- **QNH** - The station level pressure reduced to MSL in accordance with ISA conditions. When QNH set, altimeter indicates elevation on ground and altitude in flight.
- **QNE** – Standard Altitude setting 1013.25 hPa. When QNE set, altimeter will read FL/PA.

- **QFE**- The value of pressure, for a particular aerodrome and time, corrected to the official elevation. When set on ground altimeter will read zero. When set in flight, altimeter will read height above airport elevation.
- **QFF** The station level pressure reduced to MSL in accordance with prevailing atmospheric conditions.

Transition Altitude - This is the altitude at or below which the vertical position of an aircraft is expressed and controlled in terms of altitude. In India its value is commonly 4,000 feet.

Transition Level - This is the lowest flight level available for use above the transition altitude. At and above transition level, vertical position is expressed as a flight level.

Transition Layer –

- This is the airspace between transition altitude and transition level.
- When climbing through it, the aircraft's vertical position is expressed in terms of flight level.
- When descending through it, in terms of altitude.

Density altitude –

- Density altitude can be defined as the altitude in the standard atmosphere at which the prevailing density would occur.
- If density is more, Density Altitude will be lower.
- If temperature is more than ISA, Density Altitude will be more.
- If temperature is same as ISA, Density Altitude = Pressure Altitude
- **Density Altitude = Pressure Altitude (+ or - ISA deviation) $\times$ 120**
- Where ISA Deviation = Actual Temp. – ISA Temp.
- Hence, if actual temperature is more than ISA, DA will be more than PA.
- If actual temperature is less than ISA, DA will be less than PA.

Questions

1. In the International Standard Atmosphere, the mean sea level pressure is, the lapse rate of temperature between MSL and and is isothermal up to The numbers missing are:
 - a. 1225 mb; 2° per 1000 ft; 37 000 ft; 66 000 ft.
 - b. 1013.25 mb; 1.98°C per 1000 ft; 36 090 ft; 65 617 ft.
 - c. 1013.25 mb; 1.98°C per 1000 ft; 36 090 ft; 104 987 ft.
 - d. 1225 mb; 1.98°C per 1000 ft; 36 090 ft; 104 987 ft.
2. An aircraft taking off from an airfield with QNH set on the altimeter has both static vents blocked by ice. As the aircraft climbs away the altimeter will:

- a. read the airfield elevation.
 - b. indicate the aircraft height amsl.
 - c. read the height of the aircraft above the airfield.
 - d. show only a very small increase in height.
3. **When flying from low pressure to high pressure, the barometric error of an altimeter will cause the instrument to:**
- a. read the true altitude, providing a correction is made for temperature.
 - b. overread the true altitude of the aircraft.
 - c. indicate a higher altitude than the correct one.
 - d. underread the true altitude of the aircraft.
4. **The errors affecting the pressure altimeter are:**
- a. instrument position, manoeuvre induced, density, temperature, lag.
 - b. instrument, pressure, manoeuvre induced, density, temperature, lag.
 - c. instrument, position, manoeuvre induced, temperature, barometric, lag.
 - d. instrument, pressure, lag, barometric, temperature, compressibility.
5. **An altimeter with set on the subscale will indicate, but with set, the altimeter will show**
- a. 1013; pressure altitude; QNH; height above mean sea level.
 - b. QNE; pressure altitude; QNH; height above airfield datum.
 - c. QFE; height above the airfield datum; 1013; height amsl.
 - d. QNH; height above touch down; 1013; height amsl.
6. **An a/c has one altimeter set to QFE and one to aerodrome QNH 1000 mb. If the airfield elevation is 300 ft, immediately before take-off the altimeter with QFE set will read and the other If the QFE altimeter is set to 1013 when passing through the transition altitude 3000ft, it will read..... (Assume 1 mb = 30 ft).**
- a. 300 ft; zero; 2610 ft
 - b. zero; 300 ft; 3390 ft
 - c. zero; 300 ft; 3690 ft
 - d. zero; 300 ft; 2610 ft
7. **An a/c with its altimeter setting of 29.75 inches in flying at a level where atmospheric pressure is 27.55 inches. The altimeter will read?**
8. **An altimeter with barometric subscale set to 999 mbs reads 1500 feet if subscale is reset to 1006 mbs the alt will read?**
- a. 1290 ft
 - b. 1300 ft
 - c. 1710 ft

9. An a/c switches off with altimeter set to QNH 1014 mbs reading 600 ft on ground. Next morning when QNH 1002 mbs what will altimeter read. What will altimeter read when set to current OFE?
10. An a/c altimeter setting 1013mbs lands at an aerodrome [elev 300 ft] with QNH 1026 mbs. What will altimeter read on landing.
11. An a/c takes off from A elev 300 ft on QNH 1010 mbs and lands back after 4 hrs when altimeter reads 390 ft on ground. What is current QNH and QFE?
12. An a/c on FL 70 overflies a hill elevation 7000 ft when local MSL pressure is 1023 mbs. Will the a/c clear the hill if so by what margin?
13. An a/c leaves point A QNH 1000 MBS, ELEV 210 FT for aerodrome B QFE 1000 MBS, elev 300 ft. There is a hill midway with highest elevation at 5000 ft amsl. If a/c leaves A with altimeter reading 270 ft. What minimum indicated altimeter reading should a/c fly on QNH of A to clear the highest hill by 1500 ft.
14. An altimeter reads 7000 ft on 30.42. What is P.A?
 - a. 7500
 - b. 6500
 - c. 7009
15. If an aerodrome elevation is 600 ft and PA 660 ft. What is QNH and QFE?
 - a. 1015hpa/995hpa
 - b. 1011hpa/1031hpa
 - c. 1011hpa/991hpa

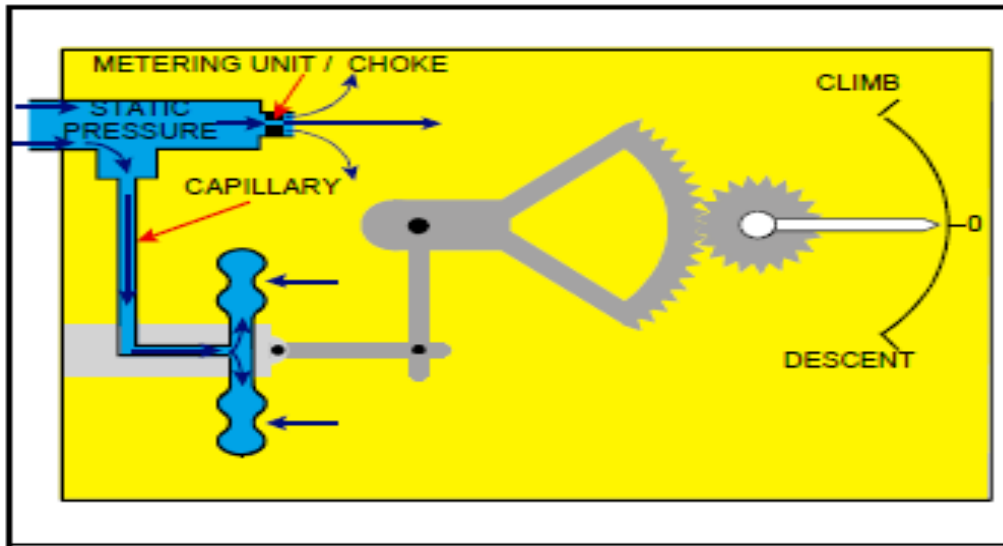
Answers

1	2	3	4	5	6	7	8
B	A	D	C	A	B	2200'	C

9	10	11	12	13	14	15
960', 0	-90'	1007,997 hpa	Yes, 292.5 ft	6410 ft	B	C

Vertical Speed Indicator

The Vertical Speed Indicator (VSI) displays rate of climb or descent. The instrument senses rate of change of static by comparing the present static pressure with a recent measurement of static.



Construction –

- A capsule within an airtight case is fed with static pressure.
- The case is also fed with static pressure but through a restricted choke/ metering unit.
- In a metering unit, air can only pass at a controlled rate.
- When aircraft climbs, the pressure in the capsule will be less than that in the case.
- The capsule will therefore contract and this compression is converted by a suitable linkage to a pointer indicating rate of climb.
- Similarly, during descent the capsule will expand and show rate of descent.

Errors of VSI -

1. Instrument Error - Due to manufacturing imperfections.

2. Position (or Pressure) Error – Due to incorrect static.

Manoeuvre-Induced Error - Any short term fluctuations in pressure at the static vent during attitude changes will cause the instrument to indicate a false rate of climb or descent.

3. Lag Error - The pointer takes a few seconds to steady because of the time taken to build up a steady pressure difference on climb or descent. There will also be a time lag on levelling out because of the time taken for the pressures to equalise. This error is most noticeable after a prolonged climb or descent, especially at a high rate.

4. Blockage - It will cause the needle to return to zero as it measures only change in static pressure.

Note - Basic VSI has inherent time lag. In high performance a/c, an instantaneous reaction is required. Hence, dashpot accelerometer is used to eliminate the lag in the IVSI.

The Instantaneous Vertical Speed Indicator –

- To overcome the problem of lag, the Instantaneous Vertical Speed Indicator (I.V.S.I) incorporates an accelerometer unit (sometimes called a dashpot or vane-type) which responds quickly to a change of altitude.
- The piston in the vertical acceleration pump immediately rises in the cylinder and causes a temporary increase of pressure in the capsule.
- The capsule expands and the pointer will give an instant indication of descent.

Errors Peculiar to the IVSI - Because of the sensitivity of the dash-pot assembly, the instrument tends to overreact to turbulent flying conditions and small fluctuations should be ignored. In a steep, level turn, the piston will tend to sink towards the bottom of the cylinder and there will **be a false indication of a climb**.

Accuracy - The instrument should read zero, or the error should be within the permissible limits

- +/- 200 feet per minute at temperatures – 20°C + 50°C
- +/- 300 feet per minute outside these temperatures.

Questions

1. During a missed approach and go-around procedure the change of aircraft attitude plus raising of the landing gear and changing of flap settings can cause

short term unpredictable errors in certain instruments. The instruments most likely to be affected in this case are:

- a. the altimeter, artificial horizon and vertical speed indicator.
 - b. the airspeed indicator, machmeter and vertical speed indicator.
 - c. the machmeter, airspeed indicator, altimeter and vertical speed indicator.
 - d. the vertical speed indicator, airspeed indicator and altimeter.
- 2. The vertical speed indicator indications may be in error for some seconds after starting or finishing a climb or descent. The error is a result of:**
- a. a combination of time lag and manoeuvre induced errors.
 - b. a combination of position error and manoeuvre induced errors.
 - c. manoeuvre induced errors only.
 - d. a combination of time lag and instrument error.
- 3. The advantage of having the VSI dial presentation in logarithmic spacing rather than in linear spacing is that:**
- a. at low rates of climb or descent the pointer movement is much larger and so is more easily read.
 - b. readings are instantaneous.
 - c. a greater range of rates of climb and descent is shown.
 - d. the internal mechanism is simplified by deletion of the calibration choke.
- 4. In the IVSI, lag error:**
- a. is overcome by feeding a sample of static pressure to the case and delaying it to the capsule.
 - b. is virtually overcome by using a special dashpot accelerometer assembly.
 - c. is overcome by the use of logarithmic presentation.
 - d. is only overcome when initiating a climb or descent.
- 5. Because the VSI measures rates of change of static pressure and not actual values of static pressure, position error:**
- a. never affects VSI indications.
 - b. may cause errors in the VSI during the take-off run.
 - c. may cause errors in VSI indications whenever airspeed is changed, if at the same time there is a change in position error.
 - d. may cause errors in VSI indications whenever airspeed is changed, even if there is no change in position error.
- 6. When entering a steep turn, an IVSI is likely to show:**
- a. no change in altitude.
 - b. a slight climb.
 - c. a slight descent.

- d. a slight descent at high airspeed only.
7. **If the static vent becomes blocked during a climb:**
- the VSI will stop at the rate of climb of the aircraft at the time of blockage.
 - the VSI will indicate a decreasing rate of climb.
 - the VSI will return to zero.
 - the VSI will indicate an increasing rate of climb.
8. **In conditions of clear air turbulence:**
- the standard VSI is more sensitive.
 - the IVSI is more sensitive.
 - both types will react the same.
 - the vertical acceleration pump will not be affected.
9. **Change of temperature as an aircraft climbs or descends:**
- will affect VSI readings whenever temperature lapse rate differs from standard conditions.
 - is compensated at the metering unit by means of a capillary and orifice.
 - has no effect on the VSI as only static pressure is used in this instrument.
 - may be allowed for by use of tables or computer.
10. **Permissible limits of accuracy of the VSI are when within a temperature range of and outside this range.**
- + 250 fpm, on the ground, -20°C to +50°C, + 300 fpm
 - + 200 fpm, at any height, -20°C to +30°C, + 300 fpm
 - + 250 fpm, at any height, -20°C to +50°C, + 300 fpm
 - + 200 fpm, on the ground, -20°C to +50°C, + 300 fpm

Answers

1	2	3	4	5	6	7	8	9	10
D	A	A	B	B	B	C	B	B	D

The Machmeter

- In high speed aircraft the machmeter is an essential instrument.

- As an aircraft approaches the local speed of sound the airflow over some parts of the fuselage or wings may be accelerated up to the speed of sound and a shock wave will form.
- A limiting Mach number must be provided in order to avoid danger associated with flight at high Mach numbers. It is known as M_{MO} and must not be exceeded.
- The speed at which airflow over some part of the aeroplane first reaches the speed of sound and a shock wave forms is called the critical Mach number, known as M_{crit} .

Machmeter –

- It is a pressure altimeter and an ASI combined in to one instrument.
- It indicates aircraft relative speed to LSS (Local Speed of Sound).
- This is particularly important at high altitudes since the speed of sound decreases with temperature and temperature decreases with altitude.
- Hence, greater the altitude lower the speed of sound.

Speed of Sound

The speed of sound is not constant but varies with air temperature. A formula for calculating the local speed of sound (LSS) is:

$$LSS = 38.95 \sqrt{T}$$

Where –

LSS is in knots,

38.95 is a constant &

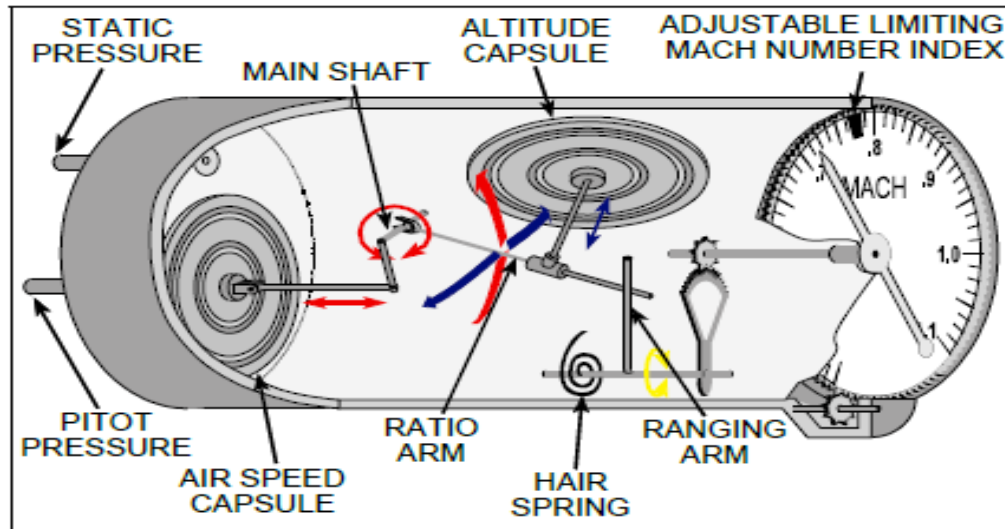
T is the absolute temperature ($0^{\circ}\text{C} = 273^{\circ}\text{K}$).

In ISA conditions at mean sea level ($+15^{\circ}\text{C}$) the speed of sound is 660 knots, while at 40,000 feet ISA (-56.5°C) the speed of sound will have reduced to 573 knots.

$\text{Mach Number} = \frac{TAS}{LSS} = \frac{P-S}{S} = \frac{\text{Dynamic Pressure}}{\text{Static Pressure}} = \frac{ASI}{ALT}$

Principle of Operation-

- The machmeter uses two capsules and linkages to indicate the aircrafts True Air Speed (TAS) as a proportion of the local speed of sound (LSS)
- The first capsule is an Airspeed Capsule which will expand and contract as a result of changes in the Dynamic pressure.
- The second capsule is a sealed Altimeter Capsule which will expand and contract as the static pressure inside the instrument case changes.



Construction –

- Aneroid altitude capsule and airspeed capsule are connected at the ratio arm.
- Static pressure enters the case of the instrument, while Pitot pressure is fed directly into the airspeed capsule.
- Expansion or contraction of both the capsules is transmitted to the ratio arm.
- If either or both capsules expand (due to an increase in IAS and / or altitude) then the ranging arm will rotate and the indicated Mach Number will increase.
- If airspeed or altitude reduce then the ratio and ranging arms move back and a lower Mach Number is displayed.
- An adjustable index on the instrument scale can be positioned by means of a small knob. This index can be set to the limiting Mach Number for the aircraft type (in straight and level flight), to provide a visual warning to the pilot.

Machmeter Errors –

- The machmeter suffers from instrument, position and manoeuvre induced errors only.
 - Instrument Error** - Like all instruments, machmeter are subject to manufacturing defects although these are extremely small.
 - Pressure Errors** – These errors are small at altitudes and speed ranges where machmeter are used.
- It does not suffer from temperature or density errors, as density changes do not alter the dynamic/static pressure ratio.
- In addition since compressibility error depends on dynamic/static pressure, and the instrument is calibrated to this ratio, compressibility error is calibrated out.
- This is the only instrument not calibrated to ISA and don't have Density, compressibility and temperature error.

- **Blockage –**

If static source blocked, Machmeter will under read in climb and over read in descent.

If pitot source blocked, Machmeter will over read in climb and under read in descent.

Accuracy of Machmeter is ± 0.1 M.

Numericals involving Machmeter –

1. What is the speed of sound at FL 380 in ISA conditions? **(573 knots)**
2. Determine the TAS corresponding to 0.70 M at MSL ISA (+15°C or 288°K). **(463 knots)**
3. If a decrease of 0.12 in the Mach Number results in a decrease of 80 Knots in the TAS, what is the local speed of sound. **(667 Kts)**

QUESTIONS

1. **The local speed of sound is equal to: (K = Constant)**
 - a. $K \sqrt{\text{temperature } (^{\circ}\text{F})}$ knots
 - b. $K \sqrt{\text{temperature } (^{\circ}\text{K})}$ knots

- c. $K \sqrt{\text{temperature } (^{\circ}\text{C. knots})}$
2. **At FL 350 with a JSA deviation of -12, the true airspeed when flying at M 0.78 is:**
 - a. 460 kt
 - b. 436 kt
 - c. 447 kt
 3. **When climbing at a constant mach number below the tropopause through an inversion:**
 - a. the CAS and TAS will both increase.
 - b. the CAS and TAS will both decrease.
 - c. the CAS will decrease and the TAS will increase.
 4. **When descending below the tropopause under normal conditions (increasing temperature) at a constant CAS:**
 - a. both TAS and mach number will decrease.
 - b. both TAS and mach number will increase.
 - c. the TAS will decrease and the mach number will increase.
 5. **Cruising at FL390, M 0.84 is found to give a TAS of 499 kt. The ISA deviation at this level will be:**
 - a. -17
 - b. +17
 - c. +19
 6. **The errors to which the machmeter is subject are:**
 - a. instrument error, position error, compressibility error and manoeuvre induced error.
 - b. instrument error, position error and manoeuvre induced error.
 - c. instrument error, position error, barometric error, temperature error and manoeuvre induced error.
 7. **The relationships between TAS, mach number (MNo) and local speed of sound (LSS) is:**
 - a. $LSS = MNo/TAS$
 - b. $MNo = LSS/TAS$
 - c. $TAS = MNo \times LSS$
 8. **The machmeter gives an indication of mach number by measuring the ratio:**
 - a. static pressure / dynamic pressure

- b. dynamic pressure / pitot pressure
 - c. dynamic pressure / static pressure
9. An aircraft is flying at FL350 with a ISA deviation of +8. The mach no is 0.83 and the TAS 485. If the aircraft descends to FL300 and maintains the same mach no and TAS, the ISA deviation will now be:
- a. +8
 - b. -2
 - c. +2
10. Determine the TAS corresponding to 0.70 M at ISA MSL (+15°C or 288K).
11. Calculate the altitude in the atmosphere at which a TAS of 450 knots corresponds to Mach 80. ($T=209^\circ$ Absolute which is equal to -64° C)
12. If a decrease of 0.12 in the Mach Number results in a decrease of 80 Knots in the TAS what is the local speed of sound?
13. An aircraft is flying at FL360 with a TAS of 467 Knots at Mach No 0.8 when the temperature difference from ISA is +9. What is the Temperature difference or FL320 if Mach 0.8 still gives a TAS of 467 Knots.
14. At flight level 330 the RAS of an aircraft is 285kt. The temperature deviation from the standard is -12° C (ISA). Use your computer to determine:
- a. The TAS
 - b. The local speed of sound
 - c. The Mach Number
15. Calculate, without using the computer, the altitude in the standard atmosphere at which 470kt TAS corresponds to Mach 0.82
- a. FL283
 - b. FL310
 - c. FL360
16. If a decrease of 0.13 in the Mach number results in a decrease of 77kt in the TAS what is the real speed of sound?
- a. 650 kts
 - b. 394 kts
 - c. 592 kts

17. An aircraft is at FL350, TAS 463kt at Mach 0.79 when the temperature deviation from standard is +9 C. Without using the computer give the temperature deviation at FL 310 which at Mach 0.79 would give a TAS of 463kt.

- a) +1°
- b) -46°
- c) -24°

Answers

1	2	3	4	5	6	7	8	9
B	B	C	A	B	B	C	C	B

10	11	12	13	14	15	16	17
470 kts	40,000'	666kts	+1°C	454kts, 564kts, .80M	C	C	A

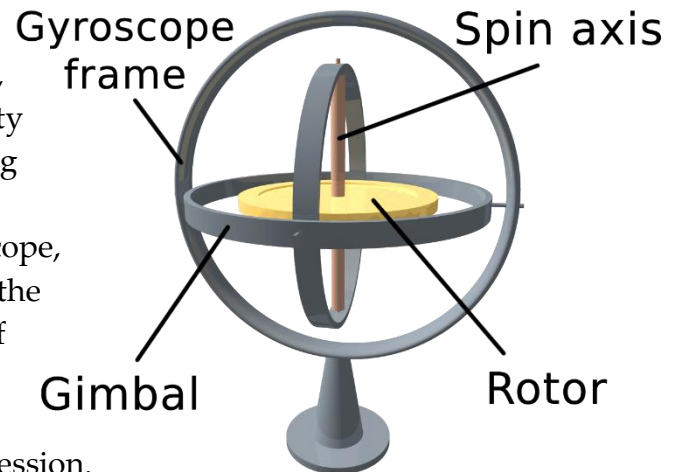
Gyroscopes

Gyroscope is a rotating body which has a mass and freedom to rotate about one or more axis. It has 2 fundamental properties:-

- i. **Rigidity in space** – A gyro will maintain its axis pointing in a fixed direction in space, unless subjected to an external force. Rigidity of a gyroscope can be increased by increasing mass of rotor, radius of rotation and RPM.
- ii. **Precession** – If a force is applied on a gyroscope, its angular momentum resists the force and the applied force acts after 90° in the direction of applied force. This angular change in the direction of spin axis is called precession.

Greater the applied force, greater is the precession.

Note – Rigidity and precession are inversely proportional to each other.



Rate of Precession - The formula for the rate of precession (Ω) is:-

$$\Omega = \frac{T}{I \omega}$$

Where: T is the applied torque

I is the moment of inertia of the rotor

ω is the angular velocity of the rotor

Type of gyro on the basis of axis –

- a. Horizontal Axis Gyro – Gyro in which rotor spins about horizontal axis.
- b. Vertical Axis Gyro – Gyro in which rotor spins about vertical axis.

Definitions -

- **Gimbal** – It is a device that supports a Gyro but allows it to move freely. Gyro is mounted on a gimbal.
- **Wander** – Deviation of gyro from its fixed datum, whether vertical or horizontal, is called wander.
- **Real Wander** – When spin axis shifts relative to space. It is caused due to turbulence, uneven mass distribution, mechanical imperfections and friction in gimbal rings.
- **Apparent wander** – It is due to the rotation of earth about its axis.
- **Drift** – When spin axis deviates in the horizontal plane (azimuth).
- **Topple** – When spin axis deviates in the vertical plane.
- **Apparent Drift** – Horizontal axis gyro suffers from apparent drift at $15 \times \sin \text{Lat/hr}$. In NH, heading will keep decreasing at this rate while in SH heading increases.

- **Apparent Topple** - Vertical axis gyro suffers from apparent topple at $15 \times \cos \text{Lat/hr}$.

Note –

- Apparent Wander occurs as despite of rotation of earth, gyro will remain aligned with North Meridian.
- Horizontal Axis Gyro – Zero drift at equator and max drift of $15^\circ/\text{hr}$ at poles.
- Vertical Axis Gyro - Zero topple at poles and max topple of $15^\circ/\text{hr}$ at equator.

Degree of Freedom – It is the number of plane about which a gyro can be processed. The degree of freedom is equal to number of gimbals or plane of freedom minus one.

Types of gyro -

1. Space or Free Gyro –

- Gyro which has freedom of plane in all 3 planes.
- It has 2 gimbals, so degree of freedom is 2.
- Such systems are used in inertial navigation systems and in a spacecraft.

2. Tied Gyro –

- It has freedom of movement in all 3 planes but one is tied.
- It has freedom of precession in 2 planes.
- It has 2 gimbals.
- They are used in DGI.

3. Earth Gyro –

- It has freedom of movement in all 3 planes but one is restricted by earth gravity.
- It has 2 gimbals, so degree of freedom is 2.
- They are used in Artificial Horizon.

4. Rate Gyro –

- It senses rate of change rather than degree of change.
- It has 1 gimbal, hence, degree of freedom is 1.
- Used in TSI.

5. Rate Integrating Gyro –

- It has single degree of freedom.
- Senses rate of movement about one axis.
- Used in INS.

Difference b/w Suction Gyros & Electric Gyros or Advantages of Electric Gyros –

- The electric rotor can be constructed with spin speed, giving greater rigidity and consequently improved stability.

- Operating RPM is more quickly reached when first switching on (No warm up time is required for Electric Gyros).
- The instrument case is sealed, for longer life and greater accuracy.

Questions

- 1. Rigidity of a gyroscope depends on:**
 - a. weight, force applied and speed of rotation.
 - b. rate of precession and the force applied.
 - c. weight, rate of precession and speed of rotation.
 - d. mass, radius of gyration and speed of rotation.
- 2. A force is applied to deflect a gyroscope. If the RPM of the gyro is then doubled the precession rate will:**
 - a. remain as before.
 - b. increase.
 - c. decrease.
 - d. cease altogether.
- 3. In gyroscopic theory the term 'topple' is defined as:**
 - a. real wander only, in the horizontal plane.
 - b. real wander only, in the vertical plane.
 - c. wander, real or apparent, in the vertical plane.
 - d. wander, real or apparent, in the horizontal plane.
- 4. A force applied to the spinning axis of a rotor is precessed:**
 - a. through 90° in the direction of spin of the rotor.
 - b. through 90° in the direction of spin of the rotor in the northern hemisphere through 90° in the opposite direction in the southern hemisphere.
 - c. through 270° in the direction of spin of the rotor.
 - d. at a rate proportional to the speed of rotation of the gyro.
- 5. In gyroscopic theory the term 'drift' is defined as:**
 - a. real wander only, in the horizontal plane.
 - b. wander, real or apparent, in the vertical plane.
 - c. apparent wander only, in the horizontal plane.
 - d. wander, real or apparent, in the horizontal plane.
- 6. Real wander of a gyro can be caused by:**
 - a. asymmetrical friction at the spinning axis.
 - b. rotation of the earth.

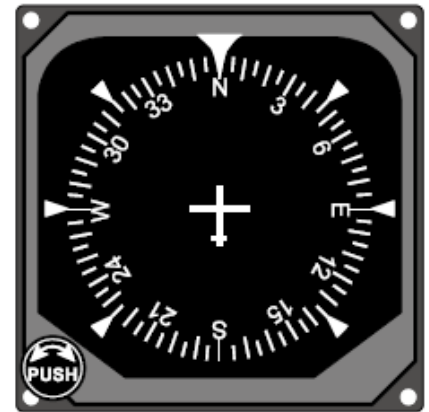
- c. increasing the RPM of the rotor.
 - d. moving the gyro north or south of its present position.
7. **A gyro with only one degree of freedom is known as a:**
- a. tied
 - b. earth gyro.
 - c. space gyro.
 - d. rate gyro.
8. **A perfectly balanced space gyro at the equator has its spin axis aligned with true north. After 6 hours the axis will be aligned with:**
- a. true east direction.
 - b. true west direction.
 - c. true north direction.
 - d. true south direction.
9. **The main advantage of electric gyros are:**
- a. light weight, high RPM, constant speed, inexpensive.
 - b. high RPM, only require low voltage DC, constant speed, sealed casing.
 - c. high RPM, high moment of inertia, rapid build-up of speed, constant RPM.
 - d. sealed casing, constant speed, high precession rate, low cost.
10. **Apparent wander of a gyro can be caused by:**
- a. rotation of the earth.
 - b. clear air turbulence.
 - c. gimbal friction.
 - d. external torque.

Answers

1	2	3	4	5	6	7	8	9	10
D	C	C	A	D	A	D	C	C	A

Directional Gyro Indicator (DGI)

- Provides Heading Reference.
- Uses Horizontal Axis Tied Gyro.
- Utilizes property of Rigidity for aircraft direction and Precession for gyro control.
- Gyro spin axis is maintained horizontal. It can be referenced to either True or Magnetic North.
- Rotor rotates at about 12,000 RPM.



Construction –

- The DI employs a tied gyro, in which gyro axis is placed in the yawing plane of the aircraft
- The rotor axis is horizontal in level flight, and because of gyroscopic rigidity it provides the datum from which heading can be measured.
- The rotor is mounted in the inner gimbal which has restricted freedom to turn while the outer gimbal can rotate through 360° about the aircraft's vertical axis.
- Initially, a/c fore and aft axis and gyro axis is aligned.
- As a/c executes a turn, the a/c fore and aft axis, instrument casing and gimbals move along with the a/c due to gyroscopic inertia or rigidity.
- But the gyro axis keeps pointing in the same fixed datum. This angular difference between a/c fore and aft axis and gyro axis is presented as Heading.
- Heading is indicated on the scale by a lubber line painted on a glass window in the instrument case.

The caging device is designed in order that:-

- The DGI can be synchronised with the compass and reset as required.
- The gyro will not topple during synchronisation.
- Toppling and possible damage to the instrument can be prevented by caging before manoeuvres in which pitch and roll limits may be exceeded.
- The DGI is reset every 15 min for accurate result. Otherwise there will be error by 4°/15minutes.

Comparison with Magnetic Compass – Magnetic compass suffers from turning and acceleration errors where as DGI does not, so tends to be more accurate especially in a steady level turn.

The Control System - Suction Gyros

- In DGI, the rotor is driven by twin jets of air applied from the outer gimbal ring.
- The jets not only spin the rotor but also serve to maintain or tie the rotor axis in the yawing plane of the aircraft.
- If the aircraft banks, outer gimbal banks with it (as it is attached to instrument case) while gyroscopic rigidity keeps the rotor axis fixed in space and they are therefore no longer 90° to each other.
- Suction air now strike one side of wedge plate more than the other.
- This applies a force to the outer gimbal that is tending to rotate it about vertical axis which is the same as applying a force to one side of the gyro rotor.
- This force is precessed by rotor to move through 90° . In direction of rotation, tilting it to keep its spin axis horizontal.
- Any tendency of the Rotor to move from the a/c horizontal reference will be corrected by wedge plate device.

DGI Limitations -

If the aircraft exceeds the pitch or roll limits of 85° (55° in air driven gyro DI's) the gyro will topple as the inner gimbal comes up against the stops, the precession causing the outer gimbal and scale to spin rapidly.

DGI Errors –

- Gimballing errors
- Random wander.
- Apparent wander due to earth's rotation.
- Errors resulting from varying rotor rpm.
- Apparent wander due to change of aircraft position (transport wander).

Gimballing errors - They occur because of the geometry of the gimbals. It is maximum during pitching and rolling simultaneously. This error disappears when turn is completed.

Random Wander – Mainly due to manufacturing imperfections.

Apparent Wander –

- The rotor axis will stay rigid in space (assuming zero real wander) while the earth rotates under it through 360° in one day or $360/24 = 15^\circ$ in one hour.
- In case of DGI, its reading will be decreasing (at the North Pole) at a rate of $15^\circ/\text{hr}$. At the South Pole the reading would increase at the same rate. This is the maximum rate of apparent wander due to the earth's rotation.
- There is no change in the direction of the rotor axis relative to the meridian, so there is zero apparent drift.
- The apparent drift rate due to the earth's rotation is therefore a function of latitude, being maximum at the pole and zero at the equator.

Hemisphere	DGI Reading's East	DGI Reading 's West
NH	Decrease	Increase
SH	Increase	Decrease

Latitude Nut –

- Provides compensation for apparent Drift.
- Fixed horizontally to the inner gimbal.
- The setting can only be made under workshop conditions (not in the aircraft)
- Latitude Nut is screwed outwards to compensate for apparent drift in NH and inwards in SH.
- It produces equal amount of real drift as of apparent drift in the opposite direction. Thereby, providing the correction.

Errors Due to Unstable RPM -

Occurs due to RPM variation because of change in altitude. As a/c climbs at high altitude suction reduces, the rotor rpm will be lower than the design value. But latitude nut continues to correct apparent drift as per the average RPM. Therefore, it overcorrects apparent drift while climbing and vice-versa while descending.

Transport Wander - At any latitude other than the equator, meridians are not parallel. If the gyro is aligned to one meridian, then flown East to West, the new meridian will be inclined to the old by Transport Wander. Sumarising:

- Flight North from the 'corrected latitude' gives a decreasing reading (a minus drift rate).
- Flight South from the 'corrected latitude' gives an increasing reading (a plus drift rate).
- Flight away from the 'corrected latitude' results in the drift rate increasing
- Flight towards the 'corrected latitude' results in the drift rate decreasing.

$$\text{Transport Wander} = \frac{\text{GS} \times \text{Tan } \theta}{60}$$

Therefore, Total Drift = Real Drift + Apparent Drift $\pm$ Transport Wander
(E = +, W= -)

QUESTIONS

1. A directional gyro indicator is basically a:

- a. horizontal axis earth gyro.
 - b. horizontal axis tied gyro.
 - c. vertical axis earth gyro.
 - d. vertical axis tied gyro.
- 2. Apparent wander may be corrected in a DGI by:**
- a. causing the gyro to precess in a clockwise direction (in the northern hemisphere).
 - b. attaching a bias weight to the inner gimbal which makes the gyro precess in azimuth in the same direction as apparent wander.
 - c. correcting wander by means of air jets.
 - d. attaching a bias weight to the inner gimbal which makes the gyro precess in azimuth in the opposite direction to apparent wander.
- 3. An air driven DGI is corrected for apparent wander at 56 N. If the aircraft is maintaining constant DGI readings:**
- a. when flying north from 56 N the true heading of the aircraft will decrease.
 - b. when flying east from 56 N the true heading will decrease.
 - c. when flying south from 56 N the true heading will decrease.
 - d. when flying west from 56 N the true heading will increase.
- 4. The formula used to calculate apparent wander of a directional gyro in the northern hemisphere is:**
- a. $+15 \sin \text{latitude in degrees for the time of running.}$
 - b. $+15 \sin \text{latitude in degrees per hour.}$
 - c. $-15 \sin \text{latitude in degrees per hour.}$
 - d. $15 \sin \text{latitude in degrees per hour increasing.}$
- 5. Errors of the directional gyro are:**
- a. acceleration error, turning error, altitude error, transport wander, rotor speed error.
 - b. gimbaling error, random wander, apparent wander, rotor speed error, transport wander.
 - c. gimbaling error, looping error, rolling error, rotor speed error, transport wander.
 - d. transport wander, apparent wander, latitude error, turning error, acceleration error.
- 6. The spin axis of a directional gyro is maintained in by means of in an air driven gyro and by means of a in an electrically driven gyro:**
- a. the horizontal plane; air jets; wedge plate.
 - b. the vertical plane; air jets; torque motor.
 - c. the yawing plane; air jets; torque motor.
 - d. the yawing plane; air jets; wedge plate.

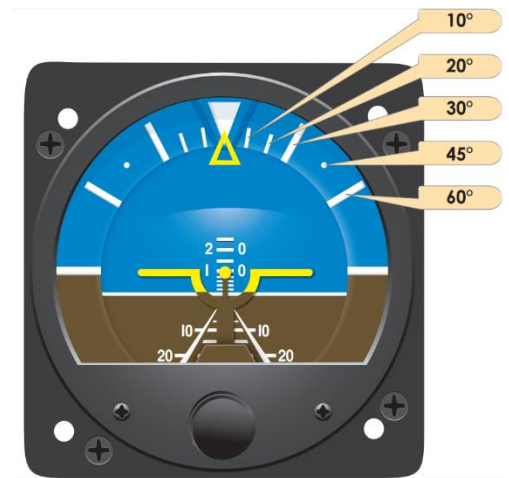
7. **The purpose of the caging knob is:**
- to prevent the gyro toppling.
 - to reset the heading.
 - to reset the heading and to prevent toppling.
 - to prevent apparent wander.
8. **In an air driven directional gyro the air jets are attached to:**
- the inner gimbal.
 - the outer gimbal.
 - the instrument casing.
 - the rotor axis.
9. **The limits of pitch and roll for a modern directional gyro are respectively:**
- 55 and 85
 - 85 and 55
 - 55 and 55
 - 85 and 85
10. **Gimballing error:**
- will disappear after a turn is completed.
 - will remain until the gyro is reset.
 - will only occur during a 360 turn.
 - will be zero on only two headings during a 360 turn.

Answers

1	2	3	4	5	6	7	8	9	10
B	D	C	C	B	C	C	B	D	A

The Artificial Horizon

- The **artificial horizon (AH)** provides the pilot with information in terms of the aircraft's attitude both in pitch and roll.
- It is a primary instrument, replacing the natural horizon in poor visibility.
- The artificial horizon uses an Earth gyro and has freedom of movement in about 3 planes.
- AH may be suction driven or electrically driven.
- RPM is 15000 for suction driven and 22000 for electrically driven.
- Uses Vertical axis earth gyro in which the spin axis is maintained in, or tied to, the vertical by Earth's gravity.



Control System –

- The control system of the air driven artificial horizon consists of four slots and four pendulous (hanging) vanes at the base of the rotor housing.
- The vanes hang down so that when the rotor axis is vertical each slot is half covered by its vane, and four equal jets of air emerge from the slots, fore an aft and left and right.
- Because the four jets are of equal strength but in opposite directions no force is exerted on the gyro and therefore no precession occurs - the gyro rotor remaining vertical.

Note – Air Driven Artificial Horizon have a speed of 8°/min. They normally take about 5 minutes to erect from startup. The rotor assembly is made slightly bottom heavy to reduce the time taken for initial erection.

Limitations -

The amount the case can move relative to the gyro is controlled by fixed stops. With older designs, typical limits are $\pm 60^\circ$ in pitch and 110° each way in roll. In modern instruments there is complete freedom in roll and up to 85° (plus or minus) in pitch.

Acceleration Error in the Air Driven Artificial Horizon -

The control system of the air driven artificial horizon depends on the pendulous vanes being affected by the Earth's gravity. However, the vanes will be affected by any acceleration, not just that due to gravity.

- Pitch Error -** During acceleration, the lateral vanes lag, swinging back towards the pilot, opening the starboard slot and closing the port slot. When an aircraft accelerates in a level attitude (such as during the take-off run) a false nose up,

right wing down, or climbing right hand turn indication will result. The pitch error is due to the effect of acceleration on the lateral pendulous vanes.

- ii. **Roll Error** - Due to inertia, the weighted base of the rotor housing tries to lag during acceleration. The roll error is due to the inertia of the bottom-heavy rotor housing.

Note - The deceleration will cause a nose down, left wing low error, the opposite of the acceleration error indication.

Turning Errors in the Air Driven Artificial Horizon –

Whenever an aircraft turns there must be an acceleration towards the center of the turn (centripetal force). Since the pendulous vanes are now affected by a horizontal acceleration as well as acceleration due to gravity, errors in pitch and roll indications will occur.

Turning Errors	Bank Angle	Pitch Attitude
Turning through 90°	Under reads bank angle	indicating a climb
Turning through 180°	Bank angle correct	indicating a climb
Turning through 270°	Over reads bank angle	indicating a climb
Turning through 360°	Bank angle correct	Pitch angle correct

Rigidity -

High rotor speeds in suction horizons of up to 15,000 rpm, result in high gyroscopic inertia. With electric horizons, speeds of 22,500 rpm are typical giving even greater rigidity.

The Electric Artificial Horizon –

- The main advantage of electric artificial horizon over the air driven horizon is its greater rigidity due to its faster spin rate of 22,000 RPM.
- This greater rigidity results in increased accuracy due to reduced errors.
- Turning and acceleration errors are eliminated.
- The basic principle of the instrument is the same as the air driven horizon. The vertical gyro is still tied by Earth's gravity, but by mercury / levelling switches and torque motors rather than the pendulous vanes of the air driven horizon.
- Mercury filled level switches are located on the inner gimbal and Torque motors on the outer gimbal.
- A cutout switch is fitted in the electrical AH to prevent it from acceleration errors when the mercury switches made contact to one end of the tube.

- There are two levelling switches, one to sense pitch and one to sense roll.
- Erection time is less than 1 minute in electric artificial horizon.
- They have a speed of 3 to 5° per minute.
- They have OFF- Flag when power supply is disconnected.

QUESTIONS

- 1. An artificial horizon utilises (i)..... to show (ii)..... in (iii)..... and (iv).....**
 - a. (i) an earth gyro (ii) position (iii) latitude (iv) longitude
 - b. (i) a space gyro (ii) attitude (iii) degrees (iv) minutes
 - c. (i) an earth gyro (ii) latitude (iii) pitch (iv) roll
 - d. (i) an earth gyro (ii) attitude (iii) pitch (iv) roll
- 2. During the take-off run an air driven artificial horizon will usually indicate:**
 - a. nose up and incorrect left bank.
 - b. a false descending turn to the right.
 - c. increased nose up attitude and right wing low.
 - d. a false climbing turn to the left.
- 3. False nose-up attitude displayed on air driven artificial horizon during the take-off run is caused by:**
 - a. the high pendulosity of the rotor
 - b. the lag of the lateral pendulous vanes
 - c. the linear acceleration cut out
 - d. incorrect rotor speed
- 4. The rotor axis of an electrical horizon is tied to the earth's vertical by:**
 - a. four pendulous vanes
 - b. the roll cut out
 - c. the low centre of gravity of the rotor housing
 - d. two mercury level switches and two torque motors
- 5. False right wing low attitude shown on an air driven artificial horizon during an acceleration is caused by:**
 - a. the lag of the base of the rotor housing
 - b. the longitudinal pendulous vanes
 - c. the roll cut-out
 - d. high rotor speed
- 6. Inside an artificial horizon:**

- a. the inner gimbal ring is pivoted laterally inside the outer gimbal ring and the outer gimbal ring is pivoted longitudinally inside the case.
 - b. the inner gimbal ring is tied to the vertical by a control system.
 - c. the rotor axis is kept level by a calibrated spring attached to the outer gimbal ring and the instrument case.
 - d. there is only one gimbal ring.
7. **When an adjustable aircraft datum is fitted to an artificial horizon in light aircraft:**
- a. it should be checked at regular intervals
 - b. it should be set to the central position and left there
 - c. it should be rendered inoperative
 - d. it should be set to 15°
8. **An electrically driven artificial horizon has less errors during the take-off run because:**
- a. it is less pendulous, has a higher rotor speed and a linear acceleration cut out
 - b. the mercury level switches are more sensitive than the pendulous vanes fitted to air driven types
 - c. the roll cut-out speed is activated
 - d. it is less aperiodic than the air driven types

Answers

1	2	3	4	5	6	7	8
D	C	B	D	A	A	B	A

The Turn and Slip Indicator

- The TSI is used to measure and display a/c rate of turn and to indicate whether the aircraft is correctly banked for no skid and no slip.
- The turn and slip are 2 different instruments, turn indicator uses gyro while slip indicator is mechanical.
- The turn indicator employs a horizontal-axis rate gyro, which has freedom of movement in one plane only i.e – Rolling/Yawing plane.
- Principle of operation is Precession for rate of turn.
- Slip indication is a separate mechanical device using a combination of Gravitational and Centrifugal Force.
- RPM used is about 4500. Low RPM is because the Principle of operation is Precession.
- Rotor Movement is up and away from the pilot.



Working –

1. When a/c is in level flight, gravity ensures the ball in center.
2. In a Co-ordinated Bank Turn, the resultant of centrifugal and gravitational force will keep ball in the center.
3. If aircraft is overbanked in a turn, centrifugal force will be less than gravitational force. Aircraft slips into a turn.
4. If aircraft is underbanked in a turn, centrifugal force will be greater than gravitational force. Aircraft will skid. Ball will move in the opposite direction.

Note - The instrument measures the precessional rate of the gyro and indicates it as a turn.

Rate of Turn –

Rate One Turn	3°/Sec
Rate Two Turn	6°/Sec
Rate Three Turn	9°/Sec
Rate Four Turn	12°/Sec

AOB for Rate 1 Turn –

10 % of TAS + 7

- a. the calibrated spring is exerting a force about the lateral axis equal to the rate of turn
 - b. the force produced by the spring is producing a precession equal to but opposite to the rate of turn is correctly banked
 - c. the spring is providing a force which produces a precession equal to the rate of turn (in the opposite direction).
 - d. the spring is providing a force which produces a precession equal to the rate of turn (in the correct direction)
4. **If the filter of the air driven rate of turn indicator becomes partially blocked:**
- a. the aircraft will turn faster than indicated
 - b. the instrument will overread
 - c. the rate of turn indicated will be unaffected
 - d. the radius of the turn will decrease
5. **The radius of a turn at rate 1, and TAS 360 kts is:**
- a. 10nm
 - b. 5nm
 - c. 7.5nm
 - d. 2nm
6. **The gimbal ring of a turn co-ordinator is inclined at about 30° with respect to the aircraft's longitudinal axis in order:**
- a. make the rate of turn more accurate
 - b. make the gyro sensitive to banking of the aircraft as well as to turning
 - c. make the gyro more effective during inverted flight
 - d. have a higher rotor speed which will prolong the life of the instrument

Answers

1	2	3	4	5	6
C	A	D	A	D	B

Terrestrial Magnetism

- Magnetic materials are 'ferrous' metals like iron and steel.
- These metals are called 'ferromagnetic' & in an aircraft they may be magnetised and produce deviation in the aircraft's compasses.
- A freely suspended magnet in the earth's Magnetic field will tend to take up an approximately North/South Direction.
- The end of bar magnet which points towards magnetic north is called Red Pole and which points towards magnetic south is called Blue Pole.
- Stroking, hammering and Magnetisation by Solenoid are some of the methods to magnetize the magnet.
- When the magnet cannot be made more powerful, it is said to be Magnetically Saturated.
- The strength of the magnet depends upon type of material and the mass of magnet.

Type of Ferrous Materials-

Type of Ferrous Materials	Metal	Ease of Magnetism	Retention of Magnetism
Hard Iron	Cobalt & Tungsten Steel	Difficult	Difficult
Soft Iron	Silicon Iron Pure Iron	Easy	Easy

Terrestrial Magnetism

- The earth behaves as though a huge permanent magnet were situated near the centre producing a magnetic field over the surface.
- The earth's blue pole lies at present beneath Northern Canada and moves approximately 1° in 5 years known as Secular Change.
- Earth's Total magnetic force is T which is the total force affecting a free compass needle.
- T can be only affected by earth's magnetic field.
- T can be resolved into two components –

Horizontal Component (H)
Vertical Component (Z)

Magnetic Dip –

- Angle that a free compass needle makes with the horizontal.
- It will be 90° at magnetic poles and 0° at magnetic equator.
- Isoclinic are the lines joining places of equal dip.
- Aclinic are the lines joining places of zero dip. Eg- Equator.

QUESTIONS

1. **The red pole of a freely suspended magnet will point towards and at latitude 60°N will point at an angle known as the angle of**
 - a. the nose of the aircraft, downwards, deviation.
 - b. the north magnetic pole, downwards, variation.
 - c. the nearest pole, downwards, declination.
 - d. the north magnetic pole, downwards, dip.
2. **If the total force of the earth's field at a point is T and the horizontal and vertical components H and Z , the value of H is found by the formula:**
 - a. $H = T \sin \text{dip}$
 - b. $H = Z \tan \text{dip}$
 - c. $H = T \cos \text{dip}$
 - d. $H = T \tan \text{dip}$
3. **The directive force of the earth's magnetic field:**
 - a. varies with the heading of the aircraft.
 - b. increases as the magnetic variation increases.
 - c. increases as magnetic latitude increases.
 - d. is greatest at the magnetic equator.
4. **The slow change in the earth's magnetic variation is known as the change and is caused by**
 - a. annual, westerly movement of the magnetic pole.
 - b. diurnal, easterly movement of the magnetic pole.
 - c. secular, westerly movement of the magnetic pole.
 - d. annual, sunspot activity.
5. **Soft iron is comparatively to magnetise whilst hard iron is to demagnetise.**
 - a. easy; difficult.
 - b. easy; easy.
 - c. difficult; easy.
 - d. difficult; difficult.

6. Which of the following materials are classed as ferromagnetic:
- iron, steel, carbon-fibre.
 - nickel, iron, steel.
 - copper, iron, carbon steel.
 - iron, cobalt steel, chromium steel.
7. The magnetic moment of a magnet:
- is the product of pole strength and effective length.
 - varies inversely as the square of the distance between the poles.
 - varies directly as the square of the distance between the poles.
 - decreases as the magnet length increases.

Answers

1	2	3	4	5	6	7
D	C	D	C	A	D	A

Direct Reading Compass

- A DRC is an instrument designed to indicate direction on the surface of the earth, relative to horizontal component of the earth's field as its directional datum.
- It is used as a main compass in small aircrafts and a mandatory requirement for big civil aircrafts as a standby compass.
- It has an accuracy of $\pm 10^\circ$.
- Near poles, the horizontal component is very small. Therefore, can't be used near poles.

There are 3 properties required for DRC –

1. **Horizontality** –
 - Magnet assembly should remain horizontal under all normal flight conditions.
 - To achieve horizontality, the magnet assembly is 'pendulously suspended', the centre of gravity of this assembly being below the pivot.
2. **Sensitivity** - Magnet assembly should be sensitive enough even for small changes in heading. Sensitivity is achieved by:-
 - By reducing the effective weight of the magnet assembly.
 - By using an iridium-tipped pivot in a jewelled cup.
 - By lubricating the pivot with the liquid which fills the compass bowl.
3. **Aperiodicity** – DRC should be aperiodic i.e- it should settle down quickly on a steady indication after being displaced by turbulence or manoeuvres. The desired aperiodicity is achieved as follows:
 - Several short magnets are used instead of one longer one.
 - Using **damping wires** to reduce oscillations.

The Compass Liquid - Various liquids, including alcohol are used. The main properties required of a compass liquid are:

- Low coefficient of expansion
- Low viscosity
- Transparency
- Low freezing point and high boiling point
- Non-corrosiveness

The compass liquid should be checked and be free from:

- ✓ Sediment
- ✓ Discolouration
- ✓ Bubbles

Errors of DRC –

Acceleration Errors –

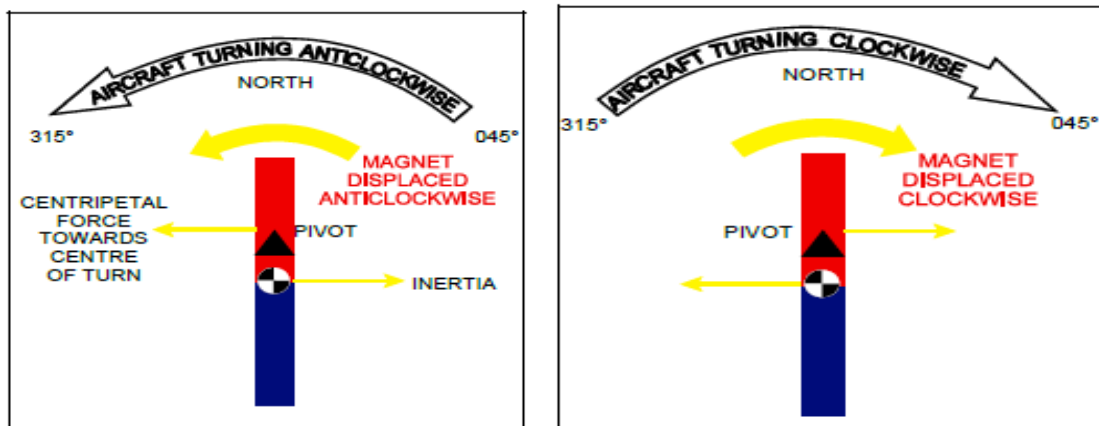
- i. Acceleration on westerly heading- (ANDS)
 - Compass card moves anti-clockwise.
 - Compass over-reads.
 - Apparent turn towards north.
- ii. Acceleration on easterly heading –
 - Compass card will rotate clockwise.
 - Compass under-reads.
 - Apparent turn towards north.

Note –

- No acceleration/deceleration errors at magnetic equator because of no dip.
- No acceleration/ deceleration errors on north-south headings.

Turning Errors –

- It is the most dominant error of DRC.
- No Turning error when turning through easterly and westerly heading if liquid swirl error is ignored.
- These are maximum when passing through magnetic North or South, decreasing to zero when passing through East or West.



- The error increases with increase in magnetic latitude.

- Whenever the pilot turns through the nearer pole:
 - ✓ the aircraft and compass rotate in the same direction,
 - ✓ the compass will be sluggish, and
 - ✓ the pilot should undershoot the turn / roll out early.
 - ✓ liquid swirl will increase the turning error.

Hemis'	Turning		Aircraft Turns	Magnets' Turn	At the end of the turn compass reads	Stop turn	Effect of liquid swirl	Compass Condition
	From	To						
N	45	315	Anti Clockwise	Anti Clockwise	More than 315 deg	Early	Increases turning error	Sluggish
N	315	45	Clockwise	Clockwise	Less than 045 deg	Early	Increases turning error	Sluggish
N	135	225	Clockwise	Anti Clockwise	More than 225 deg	Late	Reduces turning error	Lively
N	225	135	Anti Clockwise	Clockwise	Less than 135 deg	Late	Reduces turning error	Lively
S	45	315	Anti Clockwise	Clockwise	Less than 315 deg	Late	Reduces turning error	Lively
S	315	45	Clockwise	Anti Clockwise	More than 045 deg	Late	Reduces turning error	Lively
S	135	225	Clockwise	Clockwise	Less than 225 deg	Early	Increases turning error	Sluggish
S	225	135	Anti Clockwise	Anti Clockwise	More than 135 deg	Early	Increases turning error	Sluggish

Liquid Swirl Error –

- The liquid tends to swirl - and rotate the magnet assembly with it - in the same direction as the aircraft's turn. This error is called Liquid Swirl Error.
- When turning through North in NH and south in SH, Liquid Swirl Error increases and vice-versa.
- There is no turning error at magnetic equator except liquid swirl error.

Parallax Error - Occurs due to offset viewing.

Alignment Error – Occurs when Lubber Line not aligned with the compass card.

QUESTIONS

1. **In a standby direct reading compass there is:**
 - a. a non-pendulously mounted magnet system.
 - b. a single pendulously mounted bar magnet.
 - c. a circular magnet or pair of bar magnets pendulously mounted.
 - d. a low magnetic moment system, either of circular or bar configuration.
2. **The main requirements of a direct reading magnetic compass are that it should be:**
 - a. horizontal, sensitive, periodic.
 - b. easily read, floating in a transparent liquid, quick to react to change in aircraft heading.
 - c. positioned directly in front of the pilot, easily corrected for magnetic deviation, aperiodic.
 - d. aperiodic, horizontal, sensitive.
3. **For a position in the southern hemisphere, the effect of acceleration errors are greatest on headings:**
 - a. 180° and 360°
 - b. 045° and 225°
 - c. 135° and 315°
 - d. 090° and 270°
4. **An aircraft in the southern hemisphere is turning from a heading of 045° (C) to 315° (C) using a DGI. At the end of the turn the compass will read than 315° and liquid swirl will this effect.**
 - a. more; increase
 - b. less; increase
 - c. more; decrease
 - d. less; decrease
5. **In a standby compass the magnet system is immersed in a transparent liquid. The purpose of this liquid is to:**
 - a. increase sensitivity, increase aperiodicity.
 - b. increase sensitivity, decrease aperiodicity.
 - c. increase sensitivity at high latitudes, lubricate bearings.
 - d. increase sensitivity, reduce liquid swirl.
6. **To improve the horizontality of a compass, the magnet assembly is suspended from a point:**
 - a. on the centre line of the magnet.
 - b. below the centre of gravity.
 - c. above the centre of gravity.

- d. varying with magnetic latitude.
7. **The magnitude, and sense, of turning error shown by a direct reading compass varies with:**
1. the design of the compass.
 2. the direction of the turn.
 3. the rate of turn.
 4. which hemisphere the aircraft is in.
 5. the heading of the aircraft.
 6. the amount of dip at the aircraft's latitude.
- Of these statements:
- a. only 1, 2, 5 and 6 are correct.
 - b. only 1, 3, 5 and 6 are correct.
 - c. only 2, 4 and 5 are correct.
 - d. all are correct.
8. **During a sustained turn the nearer magnetic pole, the effect of liquid swirl will compass turning error.**
- a. away from; increase.
 - b. towards; not affect.
 - c. away from; not affect.
 - d. towards; increase.
9. **When carrying out a turn at the magnetic equator there will be:**
- a. no turning error.
 - b. a tendency to underread turns through south and overread turns through north.
 - c. a tendency to underread turns due to liquid swirl.
 - d. no turning error when turning through east or west

Answers

1	2	3	4	5	6	7	8	9
C	D	D	D	A	C	D	D	C

Aircraft Magnetism

Deviation –

- Deviation is the angular difference measured between the direction taken up by a Compass Needle and the Magnetic Meridian.
- Deviation is named Easterly or Westerly depending on whether the North seeking end of the compass needle lies to the East or West of the Magnetic meridian.
- It occurs due to presence of small magnetic fields created by the created by the electronic instruments inside the aircraft.

Compass Swing –

It is the process of determining deviation by comparing the aircraft's heading compass reading with magnetic heading at an isolated site.

CORRECTION OF COEFFICIENTS-

- Coefficient A** - A mechanical problem of a displaced Lubber Line corrected by aligning the lubber line with the aircrafts longitudinal axis.
- Coefficient B** – Occurs due to deviation across the longitudinal axis of the aircraft. Corrected by calculating the deviation and applying correction on Easterly or Westerly headings.
- Coefficient C** - Occurs due to deviation across the lateral axis of the aircraft. Corrected by calculating the deviation and applying correction on Northerly or Southerly headings.

$$\text{Coefficient on A} = \frac{\text{Deviation on (N+E+S+W+NE+SE+SW+NW)}}{8}$$

$$\text{Coefficient on B} = \frac{\text{Deviation on East} - \text{Deviation on West}}{2}$$

$$\text{Coefficient on C} = \frac{\text{Deviation on North} - \text{Deviation on South}}{2}$$

$$\text{Total Deviation on Heading } \theta = \text{Coefficient A} + \text{Coefficient B Sin } \theta + \text{Coefficient C Cos } \theta$$

Note – The residual deviation is recorded on a compass card and is placed on the compass.

Accuracy –

Direct Reading Magnetic Compass +/- 10°

Remote Indicating Compass +/- 1°

Occasions for Compass Swing –

- When compass components are installed or replaced.
- Whenever the accuracy of the compass is in doubt.
- After a maintenance inspection if required by the schedule.
- After a significant aircraft modification, repair or replacement involving magnetic material.
- When carrying unusual ferromagnetic payloads.
- When the compass has been subjected to significant shock.
- If the aircraft has been struck by lightning.
- After significant modification to aircraft radio/electrical systems.
- After the aircraft has been given a new theatre of operations if the move involves a large change of magnetic latitude.
- If the aircraft has been in long term storage standing on one heading.

Change of Magnetic Latitude -

Due to change in magnetic latitude, the aircraft magnetism due to hard iron component does not change but deviation does.

Due to Soft iron component, aircraft magnetism also changes along with deviation with change in magnetic latitude.

Questions

1. Compass swings should be carried out:

- a. on the apron.
- b. only on the compass swinging base or site.
- c. at the holding point.
- d. on the active runway.

2. Aircraft magnetism caused by Vertical Soft Iron:

- a. varies with magnetic heading but not with magnetic latitude.
- b. varies with magnetic latitude but not with heading.
- c. it is not affected magnetic latitude or heading.
- d. varies as the cosine of the compass heading.

3. Aircraft magnetism caused by Hard Iron:

- a. is not usually influenced by the earth's magnetic field.
- b. varies directly with magnetic latitude.
- c. varies indirectly with magnetic latitude.
- d. is maximum on east and west.

4. The aim of a compass swing is:

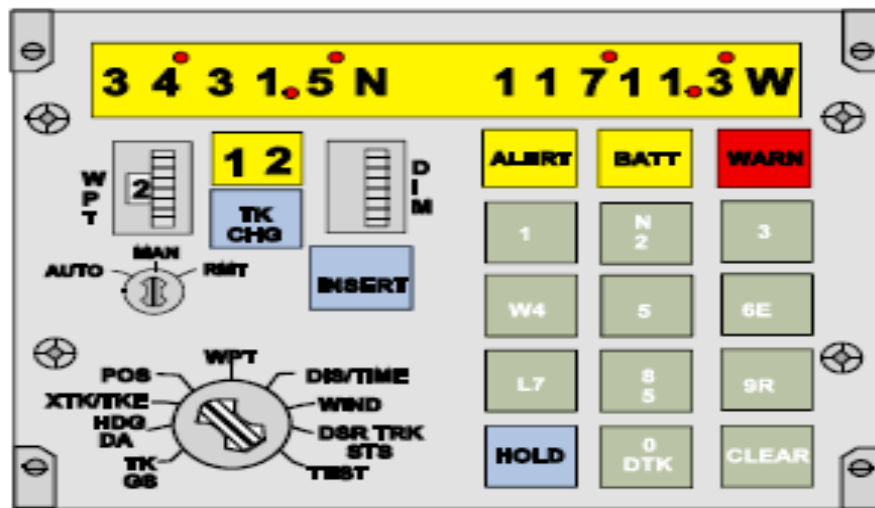
1. to find deviation on the cardinal headings and to calculate coefficients A, B and C.
 2. to eliminate or reduce the coefficients found.
 3. to record any residual deviation and to prepare a compass correction card.
- a. only answer 1 is correct.
 - b. answers 1 and 3 are correct.
 - c. answers 1, 2 and 3 are all correct.
 - d. none of the above answers are correct.
5. **Deviation due to coefficient A is mainly caused by:**
- a. hard iron force acting along the longitudinal axis.
 - b. hard and soft iron forces acting along the lateral axis.
 - c. vertical soft iron forces.
 - d. a misaligned lubber line.

Answers

1	2	3	4	5
B	B	A	C	D

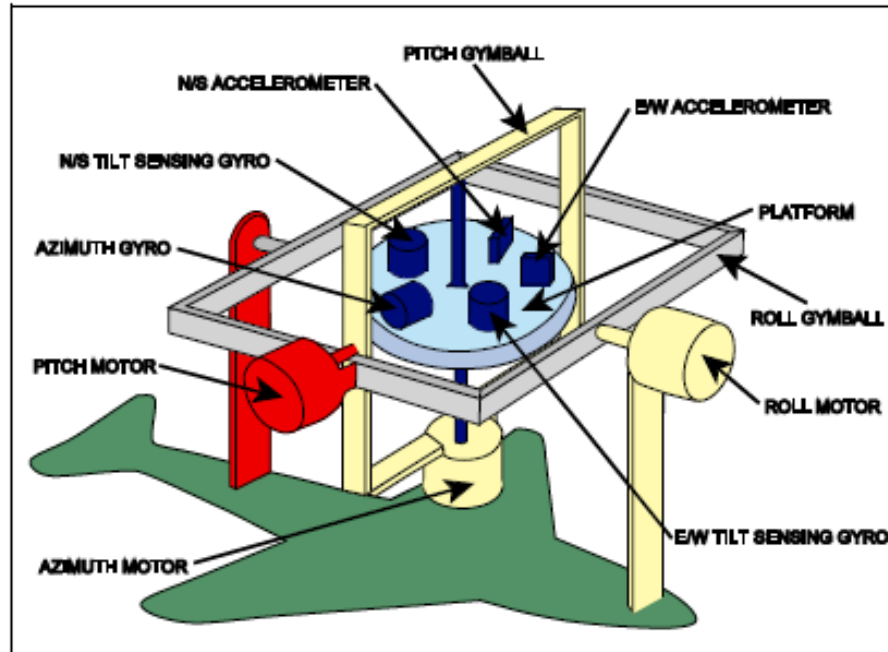
Inertial Navigation System

- INS is a computer based self-contained on board navigation system which provides velocity and position by continuously measuring and integrating its acceleration.
- It is an independent system, unaffected by weather, and can operate during the day or night.
- All corrections associated with the movement of the Earth, and transportation over the Earth's surface applies automatically.
- The product of an INS are position (Lat/Long), Speed (kt), distance (nm), and other navigational information.



Accelerometers –

- The accelerometer is basically a pendulous device.
- When the aircraft accelerates, the pendulum, due to inertia, swings off the null position.
- A signal pick off device tells how far the pendulum is off the null position. The signal from this pick off device is sent to an amplifier and current from the amplifier is sent back into a torque motor located in the accelerometer.
- A torque is generated which will restore the pendulum to the null position. The amount of current that is going into the torque is a function of the acceleration which the device is experiencing.
- The accelerometers would be mounted on a platform, there would be two, one in the North-South direction, the other in the East-West direction (often a third accelerometer is fitted to measure vertical acceleration).



- The output of accelerometers is fed to two integrators in series
 - i. Acceleration integrated with respect to time gives velocity.
 - ii. Velocity integrated with respect to time gives distance.

Errors of INS –

Bounded errors –

- Platform tilt due to initial misalignment
- Inaccurate measurement of acceleration by accelerometers
- Integrator errors in the first stage of integration.

Unbounded errors – They are either cumulative track errors or distance errors:

- Initial azimuth misalignment of the platform
- Wander of the azimuth gyro

Errors which give rise to cumulative errors in the recording of distance run:

- Wander in the levelling gyros. This causes a Schuler oscillation of the platform but the mean recorded value of distance run is increasingly divergent from the true distance run.
- Integrator errors in the second stage of integration.

Inherent Errors - The irregular shape and composition of the earth, the movement of the earth through space and other factors provide further possible sources of error. Such errors vary from system to system depending upon the balance achieved between accuracy on one hand and simplicity of design, reliability, ease of construction and cost of production.

Working-

The computer associated with the inertial system knows the latitude and longitude of the takeoff point and calculates that the aircraft has travelled so far in the North direction and so far in an East direction. The computer can then compute the new position of the aircraft and give a digital read out which we should note is to tenths of a degree.

To keep the accelerometer level, it is mounted on a gimbal assembly, commonly called the platform. The platform is nothing more than a mechanical device which allows the aircraft to go through any attitude change and yet the very inner element of the platform on which the accelerometers are mounted is able to stay earth level.

Gyroscopes which are used to stabilize the platform are also mounted on the inner-most element of the platform.

INS Mode Selector –

1. **Standby** - The power is supplied to all parts of the system. Used for warming up the system.
2. **Align** – The platform is levelled and aligned (gyro-compassed), and when these processes are complete and READY NAV illuminated.
3. **NAV** – INS makes all navigational calculations and displays them on CDU. This mode is used throughout the flight.
4. **ATT REF** – It is used when INS computer fails to provide navigation information. By selecting this mode, system will continue to provide attitude related information. (Hdg, GS etc.) All information related to navigation will be lost.
5. **Battery Light**- Illuminates in case of power supply loss. INS has its own power supply battery which supplies power for 15 min.

INS Self Alignment –

- a. **Fine leveling** - With zero output from the accelerometers fine levelling is achieved. The process takes anything up to 1 to 1½ minutes, levelling the platform to within 6 seconds of arc.
- b. **Gyro compassing** - The process of aligning the platform with local meridian. This is done by keeping the aircraft in a stationary and level state so that gyros can sense the movement of the rotating earth. This process takes up to 6 to 10 minutes, alignment accuracy of 6 seconds of arc is achieved.

Questions

1. **Gyro Compassing of an INS is accomplished in which mode?**
 - a. ALIGN
 - b. STBY
 - c. NAV
 - d. ATT REF
2. **In what way does IRS differs from INS?**
 - a. Has a shorter spin up time and suffers from laser lock.
 - b. Does not need to correct for coriolis force and acceleration.
 - c. Has a longer spin up time.
 - d. Does not experience schuler error.
3. **In a ring laser gyro, the purpose of laser lock is to –**
 - a. Compensate for transport wander.
 - b. Stabilise the laser frequency.
 - c. Enhance the accuracy of gyro at all rotational rates.
 - d. Overcome Laser Lock.
4. **What do the sensors of INS measure?**
 - a. Acceleration
 - b. Velocity
 - c. Precession
 - d. Rigidity
5. **During initialisation of an INS the aircraft must not be moved until:**
 - a. The ramp position has been inserted and checked.
 - b. The platform is levelled.
 - c. The gyros and accelerometers are in the “null” position.
 - d. The green “ready NAV” light has been illuminated and the mode selector switch has been set to the “NAV” position
6. **Dither is used in a laser gyro in order to:**
 - a. Enhance the accuracy of the gyro at all rotational rates.
 - b. Increase the maximum rotational rate that can be sensed by the gyro.
 - c. Stabilise the laser frequencies at peak power output.
 - d. Break the frequency lock which would prevent small rotational rates from being sensed by the gyro.

Answers

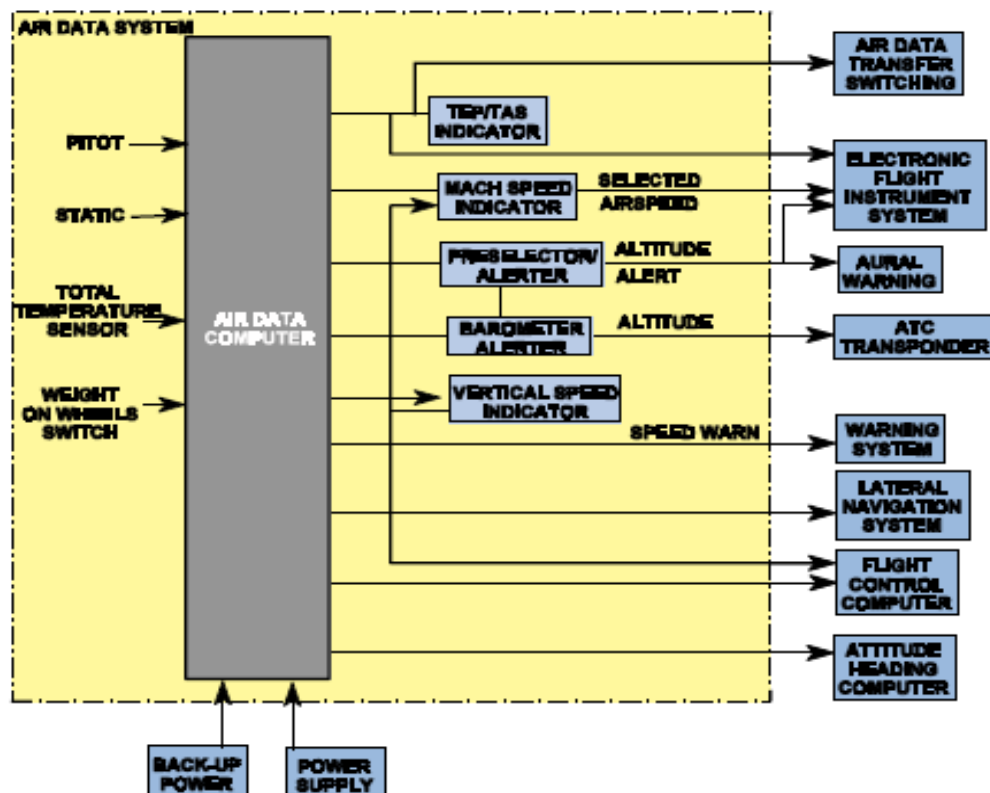
1	2	3	4	5	6
A	A	D	A	D	D

Air Data Computer

The Air Data Computer in current aircraft is a device that uses analogue or digital computing techniques to convert pressure and temperature data into electrical signals which are transmitted to the display instruments and to other systems via transducer.

In many large aircraft currently in service, the conventional pressure instruments which show altitude, airspeed and mach number are replaced by indicators displaying information generated by a central computer, The Air Data Computer (ADC).

The standard ADS instruments show altitude, vertical speed, airspeed and MNo. Additional instruments can display Total Air Temperature (TAT), Static Air Temperature (SAT) and TAS.



The output signals are used to operate various flight instruments and aeroplane systems. The flight instruments supplied by the ADC are the:-

- Altimeter
- ASI
- VSI
- Machmeter

The aeroplane systems supplied by the ADC are the:-

- Transponder
- Flight Data Recorder
- Automatic Flight Control System (AFCS)
- Ground Proximity Warning System (GPWS)
- Power Management Computer (PMC)
- Flight Director System (FDS)

These signals are computed within the ADC to produce electrical output signals, which are equivalent to –

- Altitude
- CAS
- Vertical Speed
- Mach Number
- TAS

Advantages of an Air Data System –

- Improved Displays
- Reduced Instrument and Lag Errors
- Error Correction
- Central Source for Other Systems
- Clean Design
- Failure Warning

Questions

1. An Air Data Computer (ADC)?

- a. Measures position error in the static system and transmits this information to ATC to provide correct altitude reporting.
- b. Transforms air data measurements into electric impulses driving servo motors in instruments.
- c. Is an auxiliary system that provides altitude information in the event that the static source is blocked.
- d. Converts air data measurements given by ATC from the ground in order to provide correct altitude and speed information.

2. In an ADC, airplane altitude is calculated from?

- a. The difference between absolute and dynamic pressure at the fuselage.
- b. Measurement of outside air temperature. (OAT)

- c. Measurement of elapsed time for a radio signal transmitted to the ground surface and back.
- d. Measurement of absolute barometric pressure from a static source on the fuselage.

3. The advantage provided by an ADC to indicate the altitude are:

- 1. Position/Pressure error correction.
 - 2. Hysteresis error correction.
 - 3. Remote data transmission capability
 - 4. Capability to feed data to a large number of instruments simultaneously.
- The combination of correct statement is?

- a. 1,2,3,4
- b. 2,3,4
- c. 1,2,3
- d. 1,3,4.

4. An Air Data Computer (ADC) obtains altitude from:

- a. outside air temperature
- b. barometric data from static source
- c. time elapsed for signal to travel to and return from the earth
- d. difference between absolute and dynamic pressure

Answers

1	2	3	4
B	D	D	B